# Lecture 11: Midterm Review Class

## Rational Choice and Preferences

- choices are *rationalizable* if they are consistent with always picking the best option, according to some utility function
- theorem: choices are rationalizable iff they satisfy WARP, which says: if ever you reveal $x \succsim y$, then you can never reveal $y \succ x$
- a consumer canbe desribed as maximizing utility iff he has rational *and continuous* preferences
- preferences are *convex* if averages beat extremes: if $x \equiv (x_1, x_2)$ beats $y \equiv (y_1, y_2)$, then any average of these bundles also beats $y$. Typically, test by checking for convex (up) indiff curves.

# Example

Show that lexicographic preferences over bundles $(x_1, x_2)$ are complete and transitive and convex, but not continuous.
Recall: you're only indifferent between identical bundles, and otherwise, strictly prefer whichever has more good 1; if they have the same good 1, then you prefer the one with more good 2.

# Solution

- complete: for any bundles $x = (x_1, x_2)$ and $y = (y_1, y_2)$, if one has more good 1, it's preferred; if both have the same good 1 and one has more good 2, this one is preferred; if they're identical, they're equally good. So bundles are comparable.
- transitive: if $x \succ y$, then either $x_1 > y_1$, or $x_1 = y_1$ and $x_2 > y_2$. If $y \succ z$, then either $y_1 > z_1$, or $y_1 = z_1$ and $y_2 > z_2$. If the first option applies in either case, then $x_1 > z_1$, so $x \succ z$ (the desired conclusion). If the second option applies in both cases, then $x_1 = z_1$ and $x_2 > z_2$, so $x \succ z$ (again the desired conclusion).

# Lexicographic Preferences

- convex: suppose first that you have two bundles $x$ and $y$ where $x$ has more good 1 $(x_1 > y_1)$. Then any average of them has $\alpha x_1 + (1-\alpha)y_1 > y_1$ units of good 1, thus beats $y$. Similarly if $x$ and $y$ have the same good 1 but $x$ more good 2, then weighted averages have more good 2 than $y$. So any weighted average beats $y$.
- not continuous: eg $(0,1) \succ (0,0)$, but it is not true that everything in a small enough ball around $(0,1)$ beats everything in a small enough ball around $(0,0)$ ((0,1) is strictly worse than $(\varepsilon, 0)$)

# WARP Example

- you violate WARP if in one scenario you choose bundle A when it costs weakly more (than B would have), and in another scenario you choose B when it costs weakly more (than A would have), with at least one weakly strict
- i.e. if it looks like your preferences reversed
- for example, buy bundle A on day 1 for \$100, when B would have cost \$90: this suggests $A \succ B$
- you violate WARP if you now, on another day, buy B when A would cost weakly less, eg pay \$80 for B when A would cost \$80 or less

# Marshallian Demand

- "normal" (Marshallian) demand is the best affordable bundle subject to a budget constraint
- with the intuitive approach and 2 goods, it obeys two conditions:
  - allocate money optimally between goods
  - spend all your moneh
- the optimal money allocation condition is usually $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$
  - but there are caveats: eg with perfect complements $u(x_1, x_2) = \min\{\alpha x_1, \beta x_2\}$, set $\alpha x_1 = \beta x_2$; with perfect substitutes $u(x_1, x_2) = \alpha x_1 + \beta x_2$, buy good 1 only if $\frac{\alpha}{p_1} > \frac{\beta}{p_2}$, good 2 only if the reverse inequality holds

# Compensated/Hicksian Demand

- for "compensated" (Hicksian) demand, the conditions are instead:
  - optimal money allocation (same as on the previous slide)
  - utility constraint $u(x_1, x_2) = u^*$
- expenditure is the cost of the compensated bundle, i.e. the amount of money needed to get utility $u^*$

# Example: Q1

Calculate the "usual" demand function (Marshallian demand), and also the compensated (aka Hicksian) demand function, for a consumer with utility $u(x_1, x_2) = \min\{x_1, x_2\}$. Also calculate the expenditure function.

# Answer to Q1

- for Marshallian demand, we want the best affordable bundle given prices $(p_1, p_2)$ and income $m$
- this is the solution to 2 equations:
  - optimal spending, usually it's $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$, but here it's $x_1 = x_2$
  - satisfy the **budget** constraint, $p_1 x_1 + p_2 x_2 = m$
- so,

$$x_1 = x_2 = \frac{m}{p_1 + p_2}$$

# Answer to Q1 Cont'd

- for compensated demand, we want the bundle that most efficiently, at prices $(p_1, p_2)$, achieves utility $u^*$
- this is the solution to 2 equations:
  - optimal spending, usually it's $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$, but here it's $x_1 = x_2$
  - satisfy the **utility** constraint, $u(x_1, x_2) = u^*$, here this says $\min\{x_1, x_2\} = u^*$
- so,

$$x_1^c = x_2^c = u^*$$

# Answer to Q1 Cont'd

- for the expenditure function, we want the cost of the compensated (aka Hicksian) demand bundle
- this is

$$\begin{aligned} e(p_1, p_2, u^*) &= p_1 x_1^c + p_2 x_2^c \\ &= (p_1 + p_2) u^* \end{aligned}$$

- Interpretation: eg if you want utility $u^* = 10$ at prices $(p_1, p_2) = (1, 2)$, you need \$30

# Income and Substitution Effects, and CV

- income effects ask: if the price of a good goes up, how much does demand change due to the effective wealth reduction
- substitution effects ask: if the price of a good goes up, how much does demand change due to substituting away to the other good: this is ALWAYS WEAKLY NEGATIVE (opposite direction to price change)
- note, Giffen goods must be inferior: if $p \uparrow$, the substitution effect says demand weakly falls, and if the good were normal, the income effect would also say demand falls.
- CV is the compensation needed to compensate for a price increase, so that utility is unchanged
- substitution effect is $x_i^c(p', u^o) - x_1^o$, and income effect is $x(p', m) - x_i^c(p', u^o)$

# Example: Q2

Explain: if beer and donuts are substitutes for Homer, and the price of beer increases from \$3 to \$4, and Homer currently buys 5 bottles of beer, then his compensating variation is likely less than \$5. But if they are perfect substitutes at a 1-for-1 rate, then: (i) if donuts cost more than \$4, the compensating variation if \$5; (ii) if donuts cost \$3.50, the compensating variation is \$2.50; (iii) if donuts cost \$3, the compensating variation is \$0.

# Answer to Q2

- if they are (imperfect) substitutes, then when the price of beer increases, Homer will likely respond optimally by substituting some donuts for beer. Thus even though his original bundle now costs \$5 more, he can likely get the same satisfaction with less than \$5 more money, via a different bundle with more donuts
- in (i), beer is cheapest both before/after the price increase, so Homer optimally (for perfect substitutes) buys beer only; in this case, maintaining his original satisfaction level DOES require being able to buy the same 5 bottles of beer, now \$5 more

# Answer to Q2 Cont'd

- in (ii), initially beer is cheapest, so Homer buys beer only (and we're told he chooses 5 bottles). After the price increase, donuts are cheapest, and given 1-for-1 perfect substitutes, Homer needs 5 donuts to be as happy as before. This costs $5 \cdot (\$3.50)$, whereas he originally spent $5 \cdot (\$3)$, so he needs an extra \$2.50
- in (iii), donuts and beer initially cost the same. So any combination of beer and donuts is optimal, and in particular, there would be an optimal bundle of all donuts. He could still get this when beer prices go up but donut prices do not, so he can maintain his original happiness level with no extra income.

# Lagrangians

- write out the thing you want to maximize, add any $\geq$ constraints (with a Lagrange mutliplier on each)
- optimality conditions: partial derivative of the Lagrangian should be zero with respect to all original variables you get to choose, and for each constraint, either
    - it binds: $\lambda > 0$ and the constraint holds with equality (you don't want to satisfy it, so you'll optimally *just* satisfy it)
    - it doesn't bind: $\lambda = 0$ (you can just ignore it, it's something you want to do anyway)

# Expected Utility

- so long as preferences over lotteries are complete, transitive, continuous, and satisfy the independence axiom, they can be represented by expected utility
- thus, for a lottery $p$ that yields prize $x_i$ with chance $p_i$ $(i = 1, 2, ..., n)$, the utility of the lottery is

$$EU(p) = \sum_{i=1}^{n} p_i u(x_i)$$

  where $u$ is unique up to an affine transformation
- continuity says that if $p \succ q \succ r$, you can mix $p$ and $r$ to get something better than $q$, and also mix them to get something worse than $q$

$$\begin{aligned} (0, 1, 0) &\succ (.01, .89, .1) \\ (.9, 0, .1) &\succ (.89, .11, 0). \end{aligned}$$

# Example

Sally says $(0, 1, 0) \succ (.01, .89, .1)$ and that $(.9, 0, .1) \succ (.89, .11, 0)$. Show that this is inconsistent with expected utility, and show that independence is violated. Hint: notice that the two LHS lotteries add up to the same as the two RHS lotteries.

# Solution to EU

- if EU were correct, the first preference would say

$$\begin{aligned} u(x_2) &> .01u(x_1) + .89u(x_2) + .1u(x_3) \\ &\Leftrightarrow .11u(x_2) > .01u(x_1) + .1u(x_3) \end{aligned}$$

  and the second preference would say

$$\begin{aligned} .9u(x_1) + .1u(x_3) &> .89u(x_1) + .11u(x_2) \\ &\Leftrightarrow .01u(x_1) + .1u(x_3) > .11u(x_3) \end{aligned}$$

- but these contradict each other

# Solution to Independence Axiom

- we're given $(0, 1, 0) \succ (.01, .89, .1)$ and that $(.9, 0, .1) \succ (.89, .11, 0)$
- intuitively, we guess that averaging out the two LHS lotteries should beat averaging the RHS lotteries, but this will come out to the same
- formally, by independence,

$$\begin{aligned}\frac{1}{2}(0,1,0)+\frac{1}{2}(.9,0,.1) &\succ \frac{1}{2}(.01,.89,.1)+\frac{1}{2}(.9,0,.1)\\ &\succ \frac{1}{2}(.01,.89,.1)+\frac{1}{2}(.89,.11,0)\end{aligned}$$

- a contradiction

# Applied Concepts

- a risk-averse person has $u'' \leq 0$ and $EU \leq$utility of expected wealth
- a risk-lover has $u'' \geq 0$ and $EU \geq$ utility of expected wealth
- certainty equivalent is defined by $u(w_0 + CE) = EU$ (it depends on the particular gamble): for a risk-averse person,

$$\begin{aligned} u(w_0 + CE) &= EU < u(w_0 + \text{expected wealth gain}) \\ &\Rightarrow CE < \text{expected wealth gain} \end{aligned}$$

- risk premium $\pi = E[W] - CE$ is the amount of wealth the DM will sacrifice (in expectation) to eliminate risk
- NOTE: 2023 midterm had an error! I wrongly taught them that CE was defined by $u(CE) = EU$, and so the question should have said show CE exceeds zero, not CE exceeds initial wealth

# Example

- what is your risk premium for a gamble where you are equally likely to win \$0 or \$300, assuming initial wealth \$100 and $u(w) = \sqrt{w}$?
- answer: the CE is defined by

$$\begin{aligned}
u(w_0 + CE) &= EU \text{ of gamble} \\
\sqrt{w_0 + CE} &= \frac{1}{2}\sqrt{400} + \frac{1}{2}\sqrt{100} = \frac{1}{2}(20) + \frac{1}{2}(10) = 15 \\
&\Rightarrow w_0 + CE = 225, \text{ so } CE = \$125
\end{aligned}$$

- but expected wealth gain with the gamble is $\frac{1}{2}(0) + \frac{1}{2}(300) = 150$
- so the risk premium is \$25

# Calculating Expected Utility

- this sounds simple but trips a lot of people up!!
- it is the probability-weighted sum of your possible final utilities
- if you draw a probability tree to figure this out,
    - # branches = # ways for the uncertainty to resolve
    - at the end of each branch, compute final utility if this is what happens
    - along each branch, write the chance that this branch happens
- then just sum up: add, over all branches, the probability times the associated utility

# Optimal Decisions

- an expected utility maximizer chooses the decision that maximizes expected utility

# EU Tips

- if there's a continuum of choices, **generally take a FOC**
  - if choosing $x$ to maximize EU, it is EU'(x)=0
  - caveats: a corner solution might be best for the usual reasons to want a corner solution, eg if $EU''(x) > 0$ or if EU'(x)=0 only holds at a negative $x$
- if you only have two options (eg "say yes or no to delivering the package"), compare their EU expressions and pick the best one
- if you instead differentiate EU in a parameter, you're finding whether it increases or decreases in this parameter
- caring about EU means you care about wealth and risk: if I ask for an intuitive answer, it likely means one option is better on BOTH dimensions, eg more wealth AND less risk for a risk-averse person

# Example

Suppose you decide to place a bet on the Super Bowl, which, this year, is between the Eagles and the Chiefs. You are an Eagles fan: if they lose, it's like losing \$L, with a win normalized to gaining \$0. Explain why, if you are risk-averse, you might actually bet on the Chiefs. Then characterize the optimal bet, assuming (i) if you bet $\$x$ on the Chiefs, you lose your bet if they lose, whereas if they win you get your bet back plus an additional $\frac{q}{1-q}x$. The Eagles win with chance $p$.

# Solution

- the Eagles losing is like an "accident" for you, so you might be interested in insurance where you at least get some money back in this case
- if you bet $\$x$ on the Chiefs:
  - with chance $p$, the Eagles win: in this case, you gain 0 from the win, but lose $\$x$ from your Chiefs bet
  - with chance $1 - p$, the Eagles lose: in this case, you lose $L$ from the loss, but gain $\frac{q}{1-q}x$ from your Chiefs bet
- so expected utility is

$$pu\left(w_0 - x\right) + (1-p)u\left(w_0 - L + \frac{q}{1-q}x\right)$$

# Optimal Bet

- the optimal bet solves FOC

$$-pu'(w^{\text{Eagles win}}) + (1-p)\frac{q}{1-q}u'\left(w^{\text{Eagles lose}}\right) = 0$$

- rearrange as

$$\frac{p(1-q)}{q(1-p)} = \frac{u'(w^{\text{Eagles lose}})}{u'\left(w^{\text{Eagles win}}\right)}$$

# Optimal Bet Characterization

- so if $p = q$, it's optimal to bet so that wealth is the same no matter who wins:

$$\begin{aligned} w_0 - x &= w_0 - L + \frac{q}{1-q}x \\ &\Rightarrow L = \frac{q}{1-q}x + x \\ &\Rightarrow x = (1-q)L \end{aligned}$$

- if $p > q$, $u'(w^{\text{Eagles lose}}) > u'(w^{\text{Eagles win}})$, implying wealth is higher when the Eagles win
  - so you still prefer that they win
- if $p < q$, you'll bet so much on the Chiefs that you actually want your team to lose